

Hardy's Proof of Quantum Nonlocality (Without Inequalities)

Overview

Hardy's proof demonstrates **quantum nonlocality without using Bell inequalities**. Instead of showing a numerical bound violation, it shows a **logical contradiction** between:

- **Local realism** (local hidden-variable models), and
- **Quantum mechanical predictions**

for a specific quantum state and set of measurements.

Crucially, Hardy's paradox works **only for non-maximally entangled states**.

Physical Scenario

Consider two spatially separated parties: **Alice** and **Bob**.

- Each party has two possible measurements: $x \in \{0, 1\}$ for Alice, $y \in \{0, 1\}$ for Bob.
- Each measurement has two possible outcomes: $a \in \{0, 1\}$, $b \in \{0, 1\}$.

The joint probability that Alice obtains a and Bob obtains b given measurements x and y is denoted:

$$p(a, b \mid x, y)$$

Local Realism Assumptions

1. Realism

Measurement outcomes are predetermined by hidden variables λ :

$$a = a(x, \lambda), \quad b = b(y, \lambda)$$

2. Locality

Alice's outcome does *not* depend on Bob's choice of measurement, and vice versa:

$$a(x, \lambda) \text{ independent of } y, \quad b(y, \lambda) \text{ independent of } x$$

Hardy Conditions (Logical Constraints)

Hardy identified four joint-probability conditions:

1. Nonzero “Hardy event”

$$\boxed{p(0, 0 \mid 1, 1) > 0}$$

There is a nonzero probability that both Alice and Bob obtain outcome 0 when both choose measurement 1.

2. First forbidden event

$$p(0, 1 \mid 1, 0) = 0$$

If Alice measures $x = 1$ and gets outcome 0, Bob cannot get outcome 1 when measuring $y = 0$.

3. Second forbidden event

$$p(1, 0 \mid 0, 1) = 0$$

If Bob measures $y = 1$ and gets outcome 0, Alice cannot get outcome 1 when measuring $x = 0$.

4. Third forbidden event

$$p(1, 1 \mid 0, 0) = 0$$

When both parties choose measurement 0, they never both get outcome 1.

Logical Contradiction Under Local Realism

Assume local realism and that the Hardy event occurs ($a = 0, b = 0$) for measurements $x = 1, y = 1$:

$$a(1, \lambda) = 0, \quad b(1, \lambda) = 0$$

From the forbidden events:

$$\begin{aligned} p(0, 1 \mid 1, 0) = 0 &\Rightarrow b(0, \lambda) = 0 \\ p(1, 0 \mid 0, 1) = 0 &\Rightarrow a(0, \lambda) = 0 \end{aligned}$$

Thus, for measurements $x = 0, y = 0$:

$$a(0, \lambda) = 0, \quad b(0, \lambda) = 0$$

Local realism predicts:

$$p(0, 0 \mid 0, 0) > 0$$

Quantum mechanics, however, predicts:

$$p(1, 1 \mid 0, 0) = 0 \quad \text{and} \quad p(0, 0 \mid 0, 0) = 0$$

Local realism is impossible.

Quantum Mechanical Realization

Quantum mechanics satisfies all four Hardy conditions simultaneously by:

- Choosing a **non-maximally entangled two-qubit state**
- Carefully choosing **projective measurements**

Hardy State

The two-qubit non-maximally entangled state is:

$$|\eta\rangle = \varphi^{-1}(|01\rangle + |10\rangle) + (\varphi^{1/2} - \varphi^{-1/2})|11\rangle$$

where

$$\varphi = \frac{1 + \sqrt{5}}{2}$$

Maximum Hardy Probability

The maximum probability of the Hardy event allowed by quantum mechanics is:

$$p_{\max} = p(0, 0 \mid 1, 1) = \varphi^{-5} \approx 0.09017$$

This probability is:

- Strictly positive
- Impossible under local realism
- Achieved only by non-maximally entangled states

Key Takeaways

- Hardy's paradox shows that **quantum mechanics violates local realism without inequalities**
- The contradiction arises from **logical implications**, not numerical bounds
- The paradox disappears for maximally entangled states
- The maximal Hardy probability is small but strictly nonzero